Hand-Made PCB: Non-Inverting Amplifier

Project Documentation

Course	EEE 312: Laboratory on Electronics II
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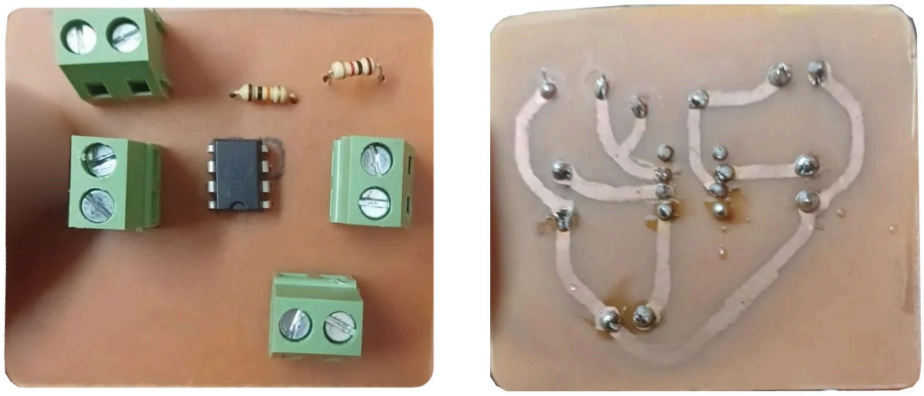
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1 Introduction

This project is a single-sided printed circuit board (PCB) for a non-inverting amplifier built around the LM741 operational amplifier. Instead of ordering the board from a manufacturer, our team of three made it entirely by hand. The layout was designed in KiCad and printed on glossy photo paper. It was then transferred to a copper-clad board with a clothes iron, etched in ferric chloride, drilled with a mini drill and soldered.

The circuit was first simulated and tested on a breadboard. The finished board was checked with a multimeter and then tested with a DC power supply, a function generator and an oscilloscope.



(a) Component side (b) Copper side

Figure 1. The finished hand-made PCB.

1.1 Project summary

Table 1. Project details

Item	Details
Project	Hand-made PCB: non-inverting amplifier
Institution	Department of EEE, University of Chittagong
Circuit	Non-inverting amplifier, LM741 op-amp, gain = 11
Board	Single-sided copper-clad board, about 58 × 50 mm
Fabrication method	Toner transfer with a clothes iron, ferric chloride etching
Software	EasyEDA (schematic), LTspice (simulation), KiCad 9 (PCB layout)
Completed	December 2025

1.2 Team

Table 2. Team members

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2 Circuit Components

2.1 Circuit

A non-inverting amplifier feeds the input signal into the **non-inverting (+) input** of the op-amp, so the output is in phase with the input. Two resistors, R1 and R2, form a voltage divider that feeds a fixed fraction of the output back to the **inverting (−) input**. This is called negative feedback, and it is what sets the gain of the circuit.

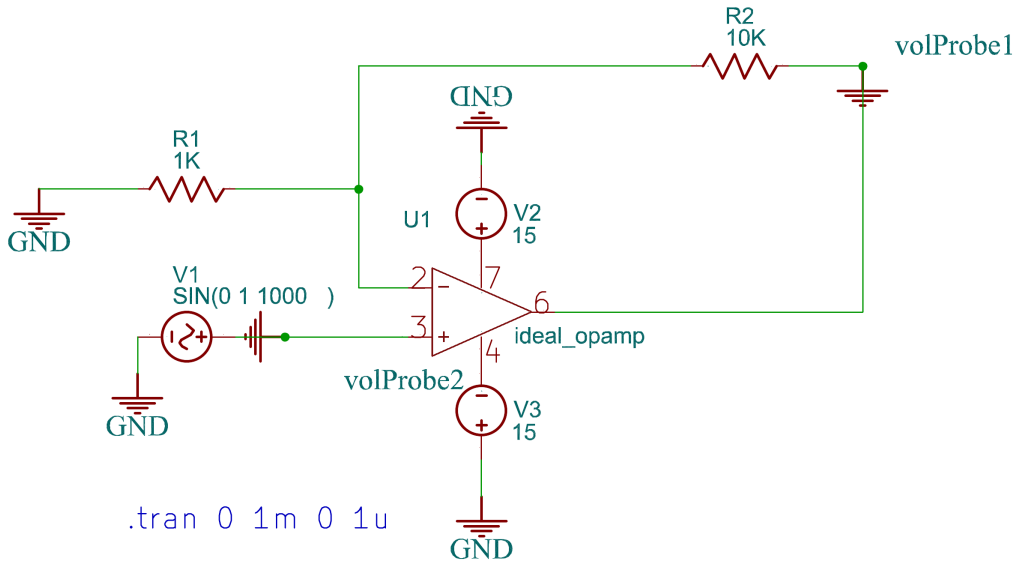


Figure 2. Circuit schematic drawn in EasyEDA. V1 is the 1 V, 1 kHz sine input; V2 and V3 are the +15 V and −15 V supplies; the volProbes mark the measured input and output.

2.2 How it works

The LM741 is an operational amplifier: an 8-pin integrated circuit that amplifies the difference between its two inputs by a very large amount (about 200 000 times in open loop). With negative feedback connected, the output settles at the value that makes the two inputs almost equal. The (−) input therefore follows the (+) input:

> $V_- = V_+ = V_{in}$
>
> V_- is set by the divider: $V_- = V_{out} \times R1 / (R1 + R2)$
>
> so $V_{out} = V_{in} \times (1 + R2 / R1)$

The input goes straight into the op-amp, which draws almost no current, so the circuit does not load the signal source. The LM741 is powered from a **split supply** (+15 V on pin 7 and −15 V on pin 4, with a common ground). This lets the output swing both above and below 0 V, which an AC signal needs. Pins 1 and 5 (offset null) and pin 8 are not used.

2.3 Component list

Table 3. Circuit components

Component	Qty	Reference	Purpose
LM741 op-amp (8-pin DIP)	1	U1	The amplifier
8-pin IC socket	1	–	Holds the IC, so the IC is not heated during soldering and can be replaced
1 kΩ resistor	1	R1	Ground resistor (– input to ground)
10 kΩ resistor	1	R2	Feedback resistor (output to – input)
2-pin screw terminal, 5.08 mm pitch	4	J1–J4	Input, –V supply, +V supply and output connections

2.4 Pin connections

Table 4. LM741 pin assignment on the PCB

Pin	LM741 function	Connected to
1	Offset null	Not connected
2	Inverting input (–)	Junction of R1 and R2
3	Non-inverting input (+)	J1 – signal input
4	Negative supply (–V)	J2
5	Offset null	Not connected
6	Output	R2 and J4 – output
7	Positive supply (+V)	J3
8	No connection	Not connected

The second pin of every screw terminal is connected to the common ground track. So each terminal carries one signal or supply line plus ground.



LM741 op-amp


8-pin IC socket


1 kΩ resistor


10 kΩ resistor


Screw terminal

2.5 Gain calculation

> Voltage gain: $A_v = 1 + R2 / R1 = 1 + 10\ k\Omega / 1\ k\Omega =$ **11**
>
> Gain in decibels: $20 \log_{10}(11) =$ **20.8 dB**
>
> For an input of 1 V peak (2 V peak-to-peak) at 1 kHz: $V_{out} = 11 \times 1\ V =$ **11 V peak** (22 V peak-to-peak), in phase with the input.

The output cannot go beyond the supply voltages. With a ±15 V supply, the LM741 output can swing to only about ±13 to ±14 V. With a gain of 11, the input should therefore stay below about 1.2 V peak. A larger input would flatten (clip) the tops of the output wave.

2.6 Simulation

Before building, the circuit was simulated in LTspice with a 1 V peak, 1 kHz sine wave at the input. The simulated output in Figure 3 is a sine wave of about 11 V peak, in phase with the input, which agrees with the calculation.

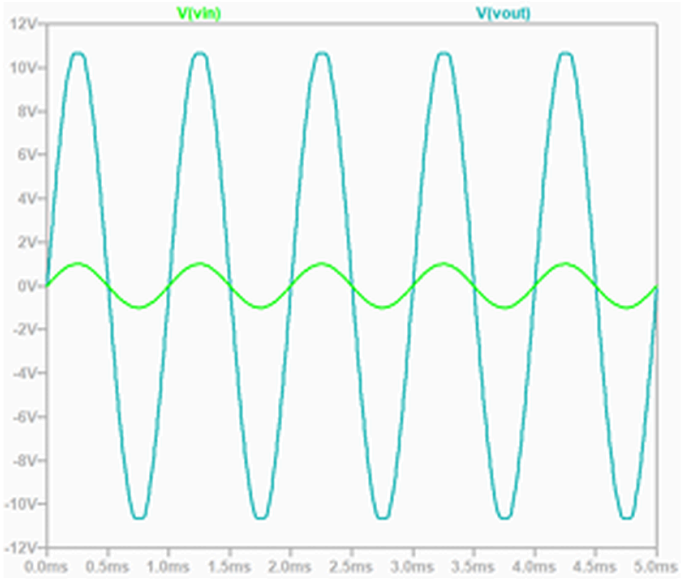


Figure 3. LTspice simulation: input V(vin) and output V(vout).

2.7 Breadboard test

Before the PCB was made, the circuit was assembled on a breadboard with jumper wires and tested with the same instruments that were later used for the board. This confirmed the connections and the gain before they were fixed in copper. A mistake is easy to correct on a breadboard but expensive once the board is etched. Figure 4 shows the LM741 with the 10 kΩ feedback resistor (brown-black-orange) and the 1 kΩ ground resistor (brown-black-red).

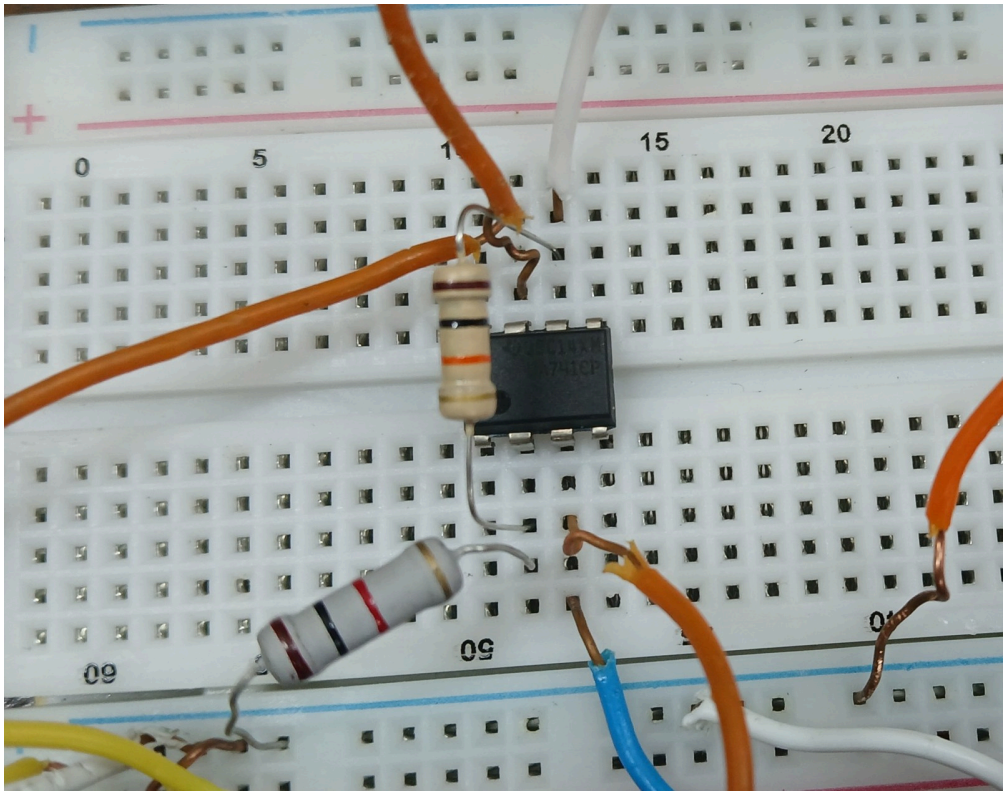
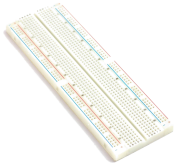


Figure 4. The circuit on the breadboard.



Breadboard



Jumper wires

3 PCB Making Materials

Table 5. PCB making materials and tools

Item	Used for
Single-sided copper-clad board	Base of the PCB
Ferric chloride ($FeCl_3$)	Etching solution that dissolves the unwanted copper
Sandpaper	Cleaning the copper before transfer; removing toner after etching
Rubber gloves	Protecting hands from the etchant
Permanent marker	Touching up broken traces before etching, if needed
Hacksaw	Cutting the board to size
Clothes iron	Heat transfer of the toner onto the copper
Mini drill machine (0.8 mm and 1.3 mm bits)	Drilling the component holes
Soldering iron, solder wire and flux	Soldering the components
Desoldering pump	Removing excess solder
Wire cutter	Trimming component leads

3.1 Materials and tools



Copper-clad board



Ferric chloride



Sandpaper



Rubber gloves



Clothes iron



Mini drill machine



Soldering iron and accessories

Figure 5. The main PCB making materials and tools.

4 Fabrication Process

The board was made in the following order. Each step is described below.

1. Design the layout in KiCad
2. Print the layout on glossy photo paper
3. Cut the copper board to size
4. Clean the copper surface
5. Transfer the toner with a clothes iron
6. Soak in water and peel off the paper
7. Etch in ferric chloride
8. Remove the toner
9. Drill the holes
10. Check continuity with a multimeter
11. Solder the components
12. Test the board

4.1 PCB layout design

The layout was drawn in KiCad 9 on a single copper layer, with all parts through-hole. The traces were kept wide (about 1.5 mm) so that they survive hand transfer and etching without breaking. The screw-terminal pads have 1.3 mm holes and the resistor and IC pads have 0.8 mm holes. The output terminal J4 was added in the final revision of the layout (Figure 6).

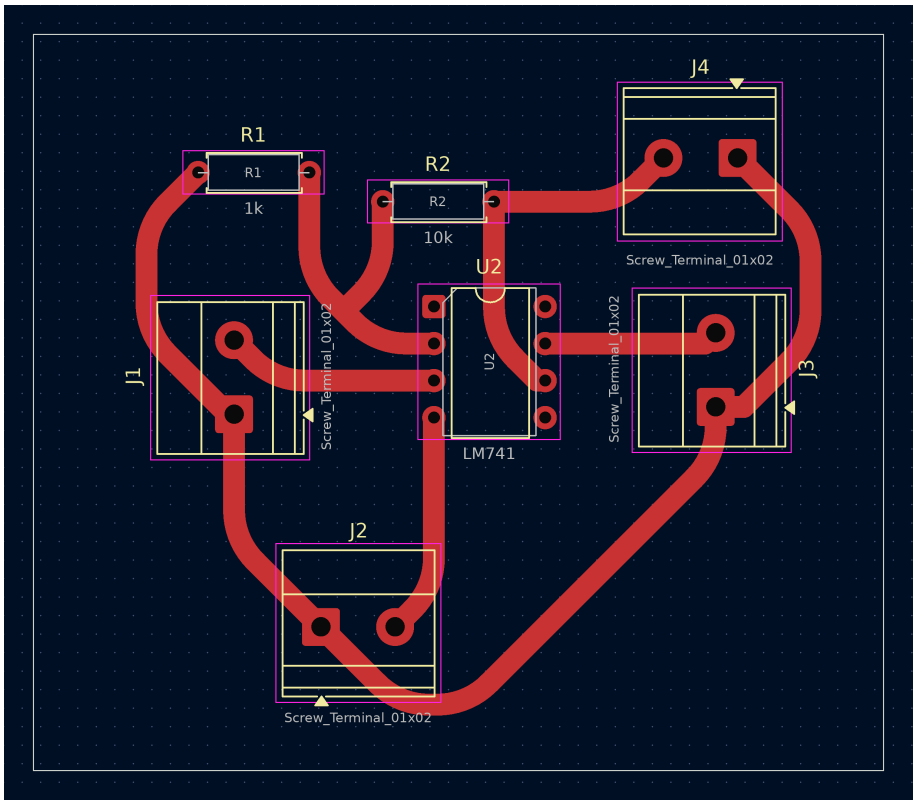


Figure 6. Final PCB layout in KiCad (top view). Red: copper traces and pads; yellow: silkscreen; pink: component courtyards.

4.2 Printing the layout

The toner-transfer method needs a **laser printer**. Laser toner is a fine plastic powder that is melted onto the paper. Under a hot iron it melts again and sticks to the copper. Inkjet ink is a liquid dye, so it would neither transfer nor protect the copper from the etchant. We did not have a laser printer, so the layout was printed at a print shop on **glossy photo paper**. Its smooth coating lets the toner release easily from the paper. Only the copper layer was printed, at 1:1 scale, with six copies on one A4 sheet so that spares were available if a transfer failed (Figure 7).

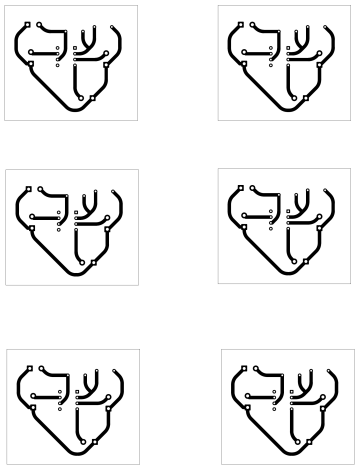


Figure 7. The copper-layer printout (six copies on one A4 sheet).

4.3 Cutting the board

The board size was marked on the copper-clad sheet and cut with a hacksaw. The cut edges were then smoothed with sandpaper so they would not lift the paper during transfer.

4.4 Cleaning the copper

Before the transfer, the copper surface was rubbed with sandpaper until it was evenly shiny, then wiped clean. Copper forms a thin oxide layer in air and picks up grease from fingers. Both stop the toner from bonding, which leads to broken traces after etching.

4.5 Toner transfer

The printout was placed toner-side down on the clean copper, and a hot clothes iron was pressed firmly on top. The iron was held there for a long time and moved slowly, so that the whole layout was heated evenly. The heat melts the toner, and the pressure bonds it to the copper.

The board was then soaked in water until the paper softened, and the paper was gently rubbed off. The toner stays on the copper as a black mask in the exact shape of the traces and pads.

4.6 Etching

Ferric chloride is an iron salt. Dissolved in water, it forms a dark brown etching solution that reacts with copper:

> $$2\,FeCl_3 + Cu \rightarrow 2\,FeCl_2 + CuCl_2$$

The copper chloride produced is green, which is why the solution turns greenish-brown as etching goes on. The toner is a plastic, so it does not react, and the copper under it is protected. Only the uncovered copper dissolves, and what remains forms the traces.

Wearing gloves, we placed the board in the solution in a plastic tray and rocked it gently until all of the exposed copper had dissolved, then rinsed it in clean water. Ferric chloride is corrosive and stains skin and clothes, and it attacks metal containers.

4.7 Removing the toner

After rinsing, the toner was scrubbed off with sandpaper. This uncovered the shiny copper traces underneath. The board was then rinsed and dried.

4.8 Drilling

A hole was drilled at the centre of every pad with a mini drill machine: 0.8 mm for the resistor and IC pins, and 1.3 mm for the screw-terminal pins. The small dot left at the centre of each pad in the layout helps keep the bit centred.

4.9 Continuity check

Before soldering, every trace was checked with a multimeter in continuity mode, which beeps when there is a connection. Each trace was tested end to end to find breaks, and neighbouring traces were tested against each other to find short circuits from copper left between them. Faults are much easier to find and fix at this stage than after the components are soldered on.

4.10 Soldering

The components were inserted from the plain side of the board and soldered on the copper side, lowest parts first: resistors, then the IC, then the screw terminals. For each joint, the tip of the soldering iron heated the pad and the lead together. Solder wire was then fed into the joint, not onto the tip, so that it flowed around the lead and formed a small cone. The flux cleans the metal surfaces so the solder can wet them. The extra lead length was then cut off with a wire cutter. The LM741 had to be mounted upside down because of a mirrored layout; this is explained in Section 6.

5 Testing

The circuit was connected as shown in Table 6. Channel 1 of the oscilloscope showed the input and channel 2 showed the output, so the two waveforms could be compared directly.

Table 6. Test connections

Terminal	Line	Connected to
J3	+V supply	DC power supply, +15 V
J2	−V supply	DC power supply, −15 V
J1	Input	Function generator (sine wave, 1 kHz) and oscilloscope channel 1
J4	Output	Oscilloscope channel 2



Figure 8. Oscilloscope reading: input on channel 1 (yellow, 1 V/div) and output on channel 2 (blue, 10 V/div).

Table 7. Expected and measured values

Quantity	Expected	Measured
Frequency	1 kHz	999.99 Hz
Input amplitude	–	2.44 V peak-to-peak (1.22 V peak)
Output amplitude	11 × 2.44 = 26.8 V peak-to-peak	26.0 V peak-to-peak (13.0 V peak)
Voltage gain	11 (20.8 dB)	26.0 / 2.44 = 10.66 (20.6 dB)
Phase shift	0°	−1.3°



DC power supply



Function generator



Oscilloscope

5.1 Discussion

The measured gain was 10.66, which is 3.1% below the calculated value of 11. The resistors have a gold tolerance band, meaning each can be up to 5% away from its marked value, so this difference is within what the component tolerances allow. Reading peak-to-peak values from the oscilloscope also adds a small error. The output stayed in phase with the input (−1.3°). The small lag comes from the LM741's limited speed: its gain-bandwidth is about 1 MHz, so with a gain of 11 the circuit's bandwidth is roughly 90 kHz, and 1 kHz is far below this. The 13 V peak output is close to the LM741's limit on a ±15 V supply, but the waveform showed no clipping.

6 Problem Faced: Mirrored Layout

While placing the components, we noticed that the board was a **mirror image** of the KiCad design. Compare the component side in Figure 1(a) with the layout in Figure 6: the output terminal J4 is at the top-left of the real board instead of the top-right, and the resistors have swapped sides.

Cause. The copper layer was printed with the mirror option turned on, and the toner transfer flipped the image a second time, because the paper is laid face down on the copper. On a single-sided board the traces are underneath and the parts are on top. So the copper, seen from underneath, has to be the mirror image of the KiCad top view. With the extra flip, it came out as the top view itself.

Effect. Resistors and two-pin screw terminals are symmetrical, so they still fitted and connected correctly. The LM741 is not symmetrical. Its pins are numbered anticlockwise when it is viewed from the top, so on a mirrored footprint every pin landed on the wrong pad.

Fix. The board had already been drilled, so we mounted the LM741 **upside down**, with its body flipped and its legs bent so that they still went into the holes. Seen from above, an upside-down IC has its pins numbered clockwise. That is exactly the mirror image, so every pin met its correct pad again.

Lesson. For toner transfer of a single-sided through-hole board, the copper side of the finished board must look like the mirror image of the KiCad top view. Checking this on the printout before ironing would have caught the mistake. In this design that means printing the copper layer without mirroring.

7 Conclusion

This project covered the complete process of making a PCB by hand, from a circuit schematic to a soldered and tested board. The non-inverting amplifier was designed for a gain of 11, simulated in LTspice and first proven on a breadboard. It was then made into a real board with the toner-transfer and ferric-chloride etching method. The measured gain of 10.66 at 1 kHz matched the design to within 3.1%.

The main practical lessons were:

- Clean, oxide-free copper is needed for the toner to stick.
- The printout must be made with laser toner, not inkjet ink.
- Heat and pressure from the iron must be even, or parts of the layout fail to transfer.
- Wide traces make hand etching much more reliable.
- Checking continuity before soldering saves a lot of time when finding faults.
- The print orientation must be checked before ironing, or the IC pins land on the wrong pads.

8 References

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